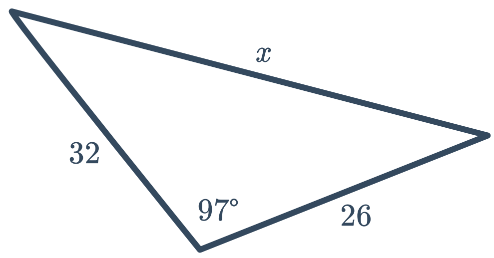
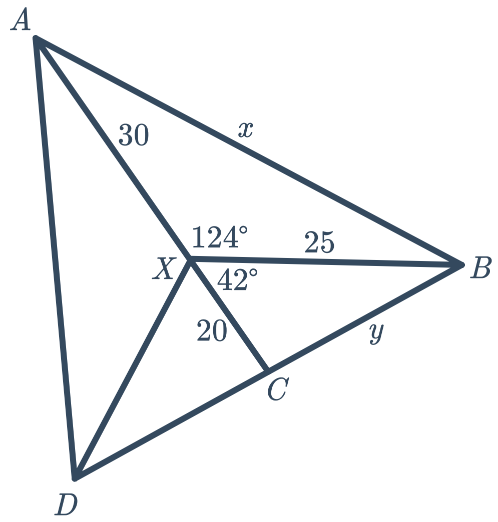
Perimeter from radial surveys

1 Consider the following triangle, taken from a field diagram. Note that all measurements are in metres.



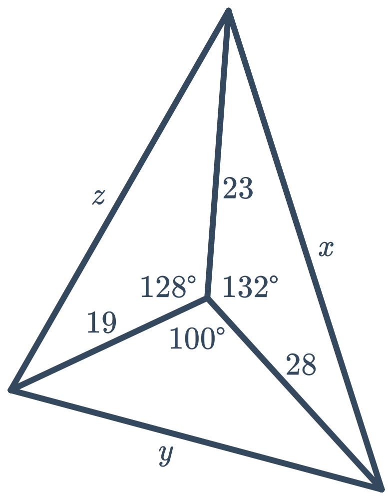
- **a** Find the side length, x, correct to two decimal places.
- **b** Hence, find the perimeter of the triangle correct to two decimal places.

2 Consider the following diagram of a field. Note that all measurements are in metres.



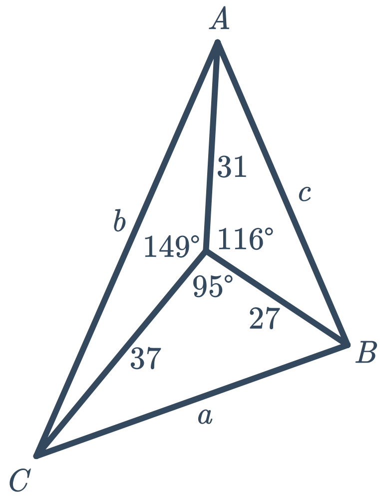
- **a** Find the value of x, correct to two decimal places.
- **b** Find the value of y, correct to two decimal places.
- **c** The lengths of CD and AD are $29.02\ \mathrm{m}$ and $44.89\ \mathrm{m}$ respectively. Find the perimeter of triangle ABD.

3 Consider the following field diagram. Note that all measurements are in metres.



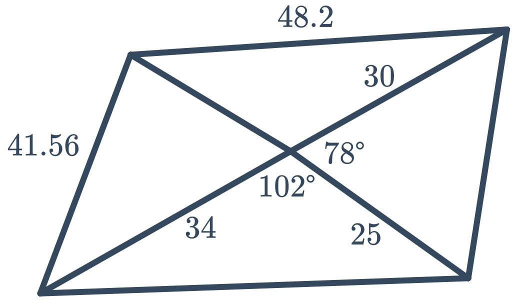
- **a** Find the value of x, correct to two decimal places.
- **b** Find the value of y, correct to two decimal places.
- **c** Find the value of z, correct to two decimal places.
- **d** Find the perimeter of the whole field correct to two decimal places.

4 Consider the following diagram. Note that all measurements are in metres.

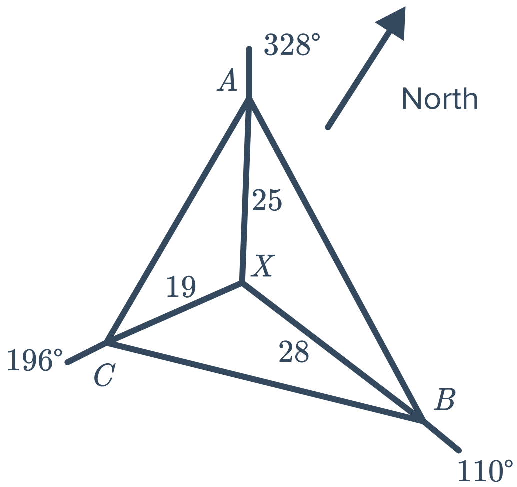


- **a** Find the value of c, correct to two decimal places.
- **b** Find the value of a, correct to two decimal places.
- **c** Find the value of b, correct to two decimal places.
- **d** Hence, find the perimeter of the field.

5 Consider the following field diagram. Find the perimeter of the whole field correct to the nearest metre.

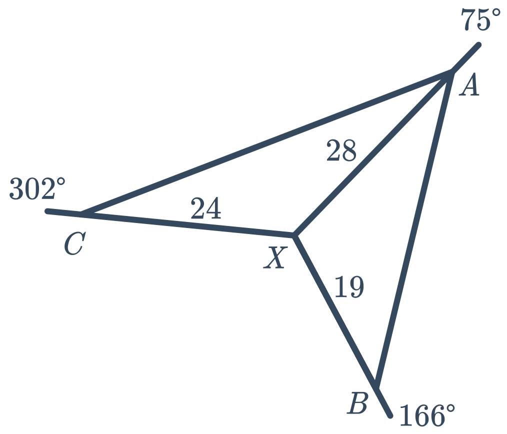


6 Consider the following diagram. Points A, B and C lie at true bearings of 328°, 110° and 196° respectively from point X.
Note that all measurements are in metres.



- **a** Find the size of $\angle AXB$.
- **b** Find the size of $\angle BXC$.
- **c** Find the size of $\angle CXA$.
- **d** Find the perimeter of the field correct to two decimal places.

7 Consider the following field diagram. Points A, B and C lie at true bearings of 75°, 166° and 302° respectively from point X. Note that all measurements are in metres.
Find the perimeter of the field diagram, correct to two decimal places.



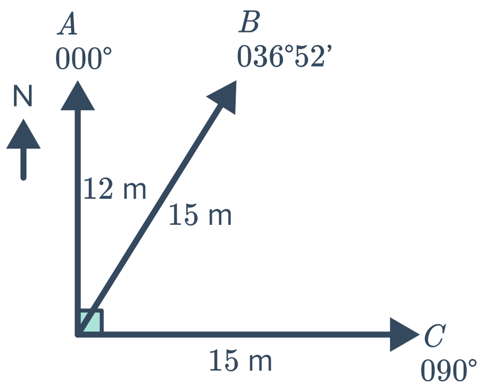
8 Consider the following radial survey of a block of land. In the survey, B is directly east of A.

a Draw a diagram of the block of land from the radial survey.

b Calculate the length of AB to the nearest metre.

c Calculate the length of BC to the nearest metre.

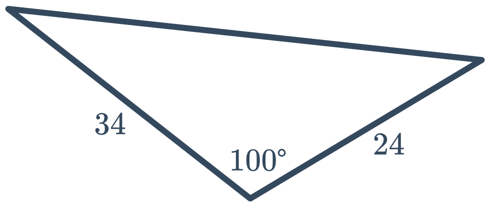
d Hence calculate the perimeter of the block of land.



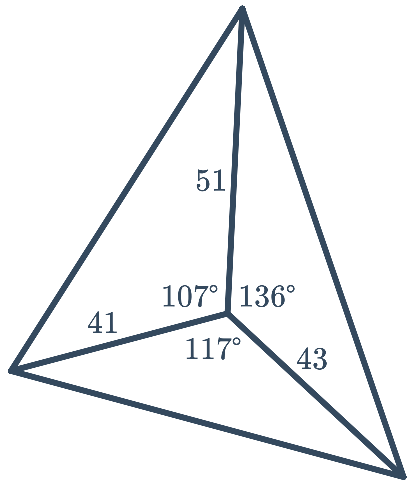
Area from radial surveys

9 Find the area of the following fields correct to two decimal places. Note that all measurements are in metres.

a



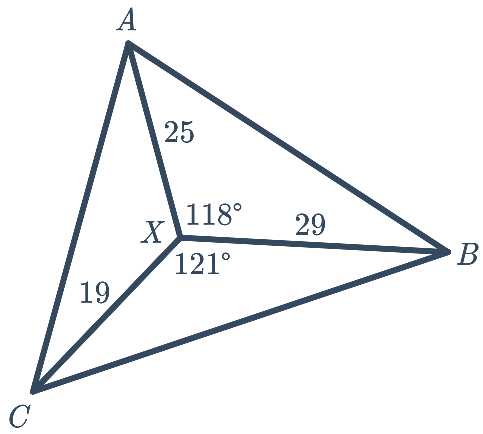
b



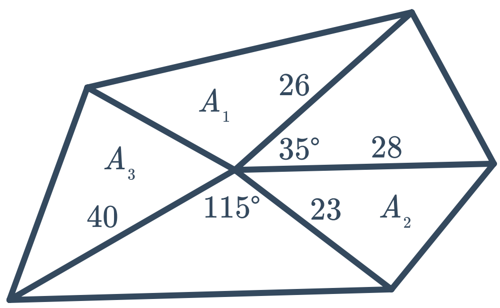
10 Consider the following field diagram. Note that all measurements are in metres.

a Find the size of $\angle AXC$.

b Find the area of the entire field, correct to two decimal places.



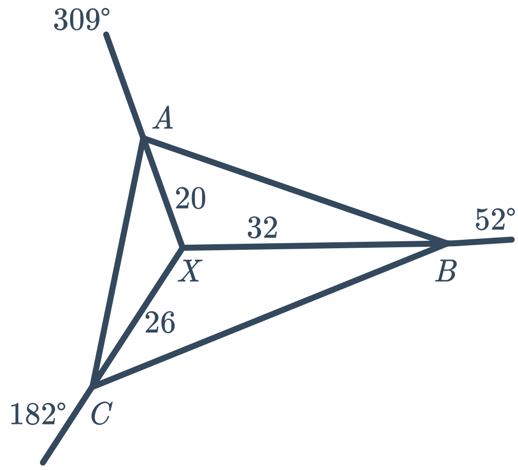
11 In the field diagram below, $A_1 = 275.23 \text{ m}^2$, $A_2 = 170.63 \text{ m}^2$ and $A_3 = 416.90 \text{ m}^2$. Find the area of the whole field correct to two decimal places.



and C lie at true bearings of 309°, 52° and 182° respectively from point X, where $AX = 20$, $BX = 32$ and $CX = 26$. Note that all measurements are in metres.

a Find the size of angles $\angle AXB$, $\angle AXC$ and $\angle CXB$.

b Find the total area of the field, correct to two decimal places.



13 Consider the following four sided field diagram. The four lines passing from the centre to the corners of the field are at true bearings of 46°, 111°, 233° and 311°. Note that all measurements are in metres.
Find the area of the field correct to two decimal places.